The allegory axioms, and what defines them in the Frobenius calculus

§1. Notation

Traditionaly Adjunction F -| G is defined as $F(x) \leq y$ iff $x \leq G(y)$, and you derive laws like this:

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) &\implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y &\implies G(F(G(y))) \leq G(y) \end{aligned}$$

By define $X : * \mapsto x$, $Y : * \mapsto y$ and use diagram order, application can be replaced by composition: $XF \leq Y$ iff $X \leq YG$

- write F as $/$, G as $\backslash$
- render $a \leq b$ as $\boxed{\frac{a}{b}}$
- **rewrite** $X \leq XF$ as $1 \leq F$

$$\boxed{\frac{X/}{Y}} = \boxed{\frac{X}{Y\backslash}} \qquad \text{adj}$$

$$\boxed{\frac{\backslash X}{Y}} = \boxed{\frac{X}{/Y}} \qquad \text{ladj}$$

$$\boxed{\frac{X/}{X/}} \xRightarrow{\{\text{adj}\}} \boxed{\frac{X}{X/\backslash}} \overset{\{\text{rewrite}\}}{=\!=\!=} \underset{\text{unit}}{\boxed{\frac{*}{/\backslash}}} \quad \begin{cases} \xRightarrow{\{\text{put } \backslash \text{ on left}\}} \boxed{\frac{\backslash}{\backslash/\backslash}} \\ \xRightarrow{\{\text{put } / \text{ on right}\}} \boxed{\frac{/}{/\backslash/}} \end{cases}$$

$$\boxed{\frac{Y\backslash}{Y\backslash}} \xRightarrow{\{\text{adj}\}} \boxed{\frac{Y\backslash/}{Y}} \overset{\{\text{rewrite}\}}{=\!=\!=} \underset{\text{counit}}{\boxed{\frac{\backslash/}{*}}} \quad \begin{cases} \xRightarrow{\{\text{put } / \text{ on left}\}} \boxed{\frac{/\backslash/}{/}} \\ \xRightarrow{\{\text{put } \backslash \text{ on right}\}} \boxed{\frac{\backslash/\backslash}{\backslash}} \end{cases}$$

now we have $/\backslash/ = /$ and $\backslash/\backslash = \backslash$, and you just discovered string diagrams.

§1.1. Every adjunction in Rel

Both halves of a row are written as sections — the operator stays and its missing argument is the gap: ·S is R ↦ R S, /S is T ↦ T/S, S\ is Y ↦ S\Y. Composition is juxtaposition, so the dot is written only in the rows where nothing else marks the gap.

F ⊣ G	F's type	𝟙⊑FG	GF⊑𝟙	F preserves ∪	G preserves ∩	FGF=F	GFG=G
◁ ⊣ ▷	A → A⊗A	𝟙⊑◁▷	▷◁⊑𝟙	(R∪S)◁= R◁∪S◁	(R∩S)▷= R▷∩S▷	◁▷◁=◁	▷◁▷=▷
⊸ ⊣ ⟜	A → 𝕀	𝟙⊑⊸⟜	⟜⊸⊑𝟙	(R∪S)⊸= R⊸∪S⊸	(R∩S)⟜= R⟜∩S⟜	⊸⟜⊸=⊸	⟜⊸⟜=⟜
° ⊣ °	(A→B) → (B→A)	R=R°°	R=R°°	(R∪S)°= R°∪S°	(R∩S)°= R°∩S°	R°°°=R°	R°°°=R°
Δ ⊣ ∩	(A→B) → $(A→B)^2$	R⊑R∩R	R∩S⊑R	—	—	R∩R=R	R∩R=R
∪ ⊣ Δ	$(A→B)^2$ → (A→B)	R⊑R∪S	R∪R⊑R	—	—	R∪R=R	R∪R=R
⊥ ⊣ !	{*} → (A→B)	—	⊥⊑R	—	—	—	—
·S ⊣ /S	(A→B) → (A→C)	R⊑(RS)/S	(T/S)S⊑T	(R∪T)S= RS∪TS	(R∩T)/S= R/S∩T/S	((RS)/S)S =RS	((T/S)S)/S =T/S
S· ⊣ S\	(B→C) → (A→C)	R⊑S\(SR)	S(S\T)⊑T	S(R∪T)= SR∪ST	S\(R∩T)= S\R∩S\T	S(S\(SR)) =SR	S\(S(S\T)) =S\T
R∩ ⊣ R⇒	(A→B) → (A→B)	X⊑R⇒(X∩R)	R∩(R⇒Y)⊑Y	R∩(X∪Y)= (R∩X)∪(R∩Y)	R⇒(X∩Y)= (R⇒X)∩(R⇒Y)	R∩(R⇒(X∩R)) =X∩R	R⇒(R∩(R⇒Y)) =R⇒Y
𝓓 ⊣ ·⊤	(A→B) → Cor A	R⊑(𝓓R)⊤	𝓓(A⊤)⊑A	𝓓(R∪S)= 𝓓R∪𝓓S	(A∩B)⊤= A⊤∩B⊤	𝓓((𝓓R)⊤) =𝓓R	(𝓓(A⊤))⊤ =A⊤
𝓡 ⊣ ⊤·	(A→B) → Cor B	R⊑⊤(𝓡R)	𝓡(⊤A)⊑A	𝓡(R∪S)= 𝓡R∪𝓡S	⊤(A∩B)= ⊤A∩⊤B	𝓡(⊤(𝓡R)) =𝓡R	⊤(𝓡(⊤A)) =⊤A
·f ⊣ ·f°	(A→B) → (A→C)	𝟙⊑ff°	f°f⊑𝟙	(R∪S)f= Rf∪Sf	(R∩S)f°= Rf°∩Sf°	ff°f=f	f°ff°=f°
f°· ⊣ f·	(A→C) → (B→C)	𝟙⊑ff°	f°f⊑𝟙	f°(X∪Y)= f°X∪f°Y	f(X∩Y)= fX∩fY	ff°f=f	f°ff°=f°
i ⊣ P	Map ↪ Rel	{·}:A→P A	∋:P B→B	—	—	Λ(∋)=𝟙	Λ(R)∋=R

S : B → C in the ·S row and S : A → B in the S· one; f is a map, B → C in the ·f row and A → B in the f°· one. Cor A is the poset of coreflexives on A: the A and B of the 𝓓 row live in Cor A, those of the 𝓡 row in Cor B — coreflexives, not objects.

§1.2. Fixed points

F ⊣ G	closed: XFG=X	open: YGF=Y
◁ ⊣ ▷	X◁▷=X	Y▷◁=Y
⊸ ⊣ ⟜	X⊸⟜=X	Y⟜⊸=Y
° ⊣ °	every X	every Y
Δ ⊣ ∩	every X	$Y_1=Y_2$
∪ ⊣ Δ	$X_1=X_2$	every Y
⊥ ⊣ !	—	Y=⊥
·S ⊣ /S	(XS)/S=X	(Y/S)S=Y
S· ⊣ S\	S\(SX)=X	S(S\Y)=Y
R∩ ⊣ R⇒	R⇒(X∩R)=X	R∩(R⇒Y)=Y
𝓓 ⊣ ·⊤	(𝓓X)⊤=X	𝓓(A⊤)=A
𝓡 ⊣ ⊤·	⊤(𝓡X)=X	𝓡(⊤A)=A
·f ⊣ ·f°	Xff°=X	Yf°f=Y
f°· ⊣ f·	ff°X=X	f°fY=Y
i ⊣ P	—	—

F and G restrict to mutually inverse isomorphisms between the two columns.

§1.3. §2.314, and what composing rows gives

One more adjunction first, which §1.1 cannot hold because it is ANTITONE — its left adjoint turns joins into meets, so that table's F preserves ∪ column would read the wrong way:

`F ⊣ G`	**the law**	**both units**	**Lean**
`R/ ⊣ \R`	`T⊑R/S` iff `S⊑T\R`	`S⊑(R/S)\R`, `T⊑R/(T\R)`	`le_div_iff` then `le_leftDiv_iff`

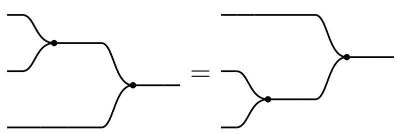
Everything below is a row of §1.1 read at a chosen argument, or two rows composed.

what you get	**which move**	**Lean**
`(R/S)(S/W)⊑R/W` [2.314]	counit of `·W ⊣ /W` at `S`, then counit of `·S ⊣ /S` at `R`	`div_comp`
`𝟙⊑R/R` [2.314]	unit of `·R ⊣ /R` at `𝟙`	`one_le_div_self`
`(R/R)(R/R)=R/R` [2.314]	`⊑` is the first line at `S=W=R`, `⊒` the second	`div_self_comp_self`
`(R/R)R=R` [2.314]	`⊑` is the counit at `R`, `⊒` is the second line composed with `R`	`div_self_comp`
`R/𝟙=R` [2.314]	`·𝟙 ⊣ /𝟙` is the identity adjunction	`div_one`
`R/(S∪T)=R/S∩R/T` [2.314]	the antitone row above: a join in the divisor turns into a meet	`div_union`
`R/(ST)=(R/T)/S`	compose `·S ⊣ /S` with `·T ⊣ /T`, then uniqueness	`div_comp_assoc`, `postDiv_comp`
`R/(fS)=(R/S)f°`	compose `·f ⊣ ·f°` with `·S ⊣ /S`, then uniqueness	`div_comp_recip_map`
`f(R/S)=(fR)/S`	compose `f°· ⊣ f·` with `·S ⊣ /S`, then uniqueness	`map_comp_div`
`(fg)°=g°f°`	compose `·f ⊣ ·f°` with `·g ⊣ ·g°`, then uniqueness	`recip_comp`, an axiom
`(R∩S)⇒X=R⇒(S⇒X)`	compose the two Heyting rows, then uniqueness	—

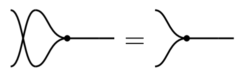
Composing is `Freyd.Adj.comp`, uniqueness is `Freyd.Adj.right_unique`, both in `Freyd/S2_313.lean`. Right adjoints compose the other way round, which is where the reversal in the first two composite lines comes from.

§2. Relations

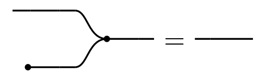
> Rel is a poset-enriched category with (⊗, °) where ⊗ is commutative cartisian product, and converse `°` ⊣ `°`, and
>
> C: `(▷ : A ⊗ A → A, ⟜ : 𝕀 → A)` a commutative monoid with `C°` ⊣ `C`. We write `C°` as `(◁ : A → A ⊗ A, ⊸ : A → 𝕀)`



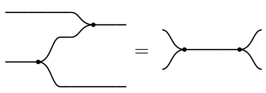
▷ associative



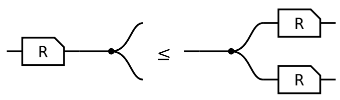
▷ commutative



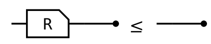
⟜ is its unit



Frobenius, one half — the other is its °

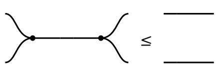


R◁ ≤ ◁(R⊗R)

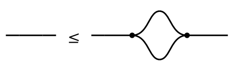


R⊸ ≤ ⊸

§2.1. ∀ object A, `(A, ◁, ⊸)` ⊣ `(A, ▷, ⟜)`



▷ ◁ ≤ 𝟙



𝟙 ≤ ◁ ▷

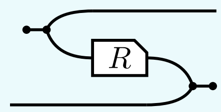
⟜ ⊸ ≤ 𝟙

𝟙 ≤ ⊸ ⟜

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In `Rel` it reads `{(a, a)} ⊆ A × A` — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§3. $\circ : \mathcal{C}^{\mathrm{op}} \longrightarrow \mathcal{C}$ is a 2 functor

° is primitive, part of the data the first section lists. It turns both of `R`'s wires round, and the Frobenius structure DRAWS that — the picture and the formula below are `R°`, not its definition:



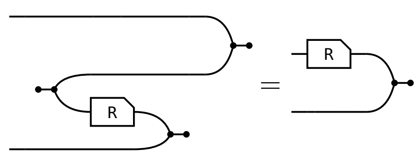
`R° = (⟜◁ ⊗ 𝟙) (𝟙 ⊗ R ⊗ 𝟙) (𝟙 ⊗ ▷⊸)`

where `⟜◁ : 𝕀 → a ⊗ a` opens a pair of wires out of nothing and `▷⊸ : b ⊗ b → 𝕀` closes one, so the input of `R°` is where the output of `R` was. Taking that formula as the definition would now be circular: ◁ and ⊸ are ▷° and ⟜°. The laws of ° are that it is a contravariant 2-functor `° : 𝒞ᵒᵖ → 𝒞`:

(i) `𝟙° = 𝟙` (ii) `(R S)° = S° R°` (iii) `(R ⊗ S)° = R° ⊗ S°` (iv) `R ≤ S` implies `R° ≤ S°`

§3.1. The slide

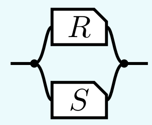
The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:



`R°` is DEFINED as the bending of `(R ⊗ 𝟙) ▷⊸`, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the ⟜◁ bends the `a` strand down and the ▷⊸ brings it back up, while the `b` strand rides through untouched. Nothing else is spent below.

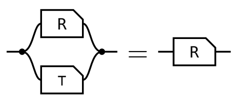
§4. ∩ is a commutative idempotent monoid on every hom-set

The **meet** of `R, S : a → b`, the paper's **convolution**, is `R ∩ S := ◁ (R ⊗ S) ▷` — copy the input, run `R` and `S` on the two copies, merge the results — so what comes out is what both of them do.

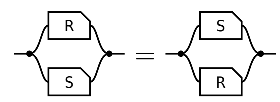


transcribed: a definition has no statement to export

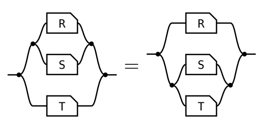
On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow `⊤ = ⊸ ⟜` (the paper's Lemma 4.11).



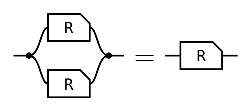
unit: one half of T per end — the merge's unit law absorbs the ⟜, the copy's counit law the ⊸



commutative: σ crosses R ⊗ S by naturality and is absorbed by cocommutativity and commutativity



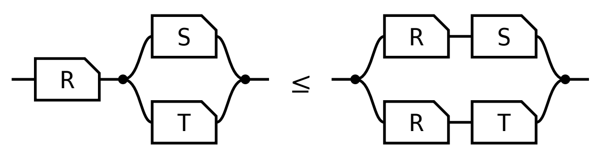
associative: coassociativity and associativity; ⊗ re-brackets for nothing, being strict here



idempotent: the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

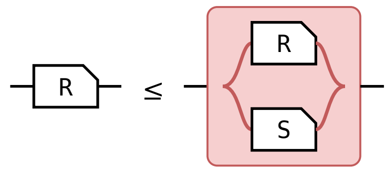
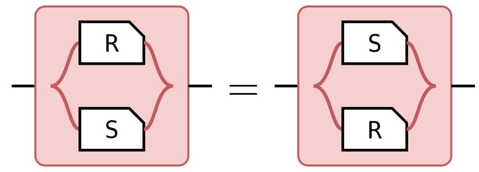
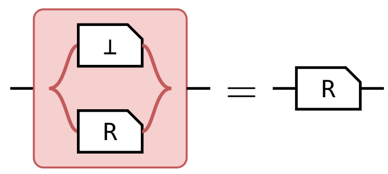
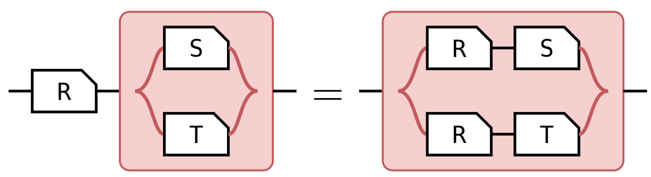
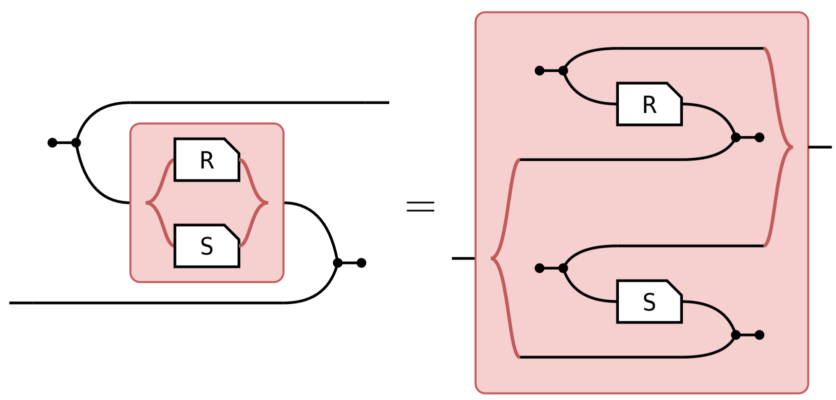
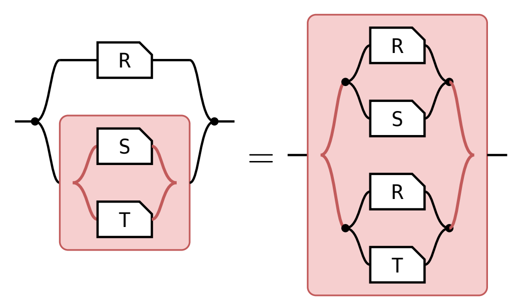
So ≤ is the order this monoid induces. `R ∩ S ≤ R` comes from the unit, and idempotency turns anything under both `S` and `T` into something under `S ∩ T`, since `R = R ∩ R ≤ S ∩ T`.

And one law relating ∩ to composition, which is **not** an equation:

semi-distributivity, and what supplies it	picture
`R (S ∩ T) ⊑ R S ∩ R T` — the lax copy law. Equality exactly when R is single valued: the Maps section's `F (R ∩ S) = F R ∩ F S`.	

§5. Union

Past what the definition can build: a union needs a **biproduct** on top of the Frobenius structure. It draws as a **tape** — the rounded wrapper is the second product, and its fork and join open and close a branch a particle takes exactly one of. The residual, which needs a second composition with linear adjoints, is in the division section below.

statement	picture
`R ⊑ R ∪ S` `union R S := ⟨R, S⟩ [𝟙, 𝟙]`: offer both branches, then forget which was taken.	
`R ∪ S = S ∪ R`	
`⊥ ∪ R = R` ⊥ routes through the zero object; there is no shape for it.	
`R (S ∪ T) = R S ∪ R T` — composition distributes over union, which `∩` does not.	
`(R ∪ S)° = R° ∪ S°`	
`R ∩ (S ∪ T) = (R ∩ S) ∪ (R ∩ T)` — what makes the whole **distributive**, and the one picture with both layers in it.	

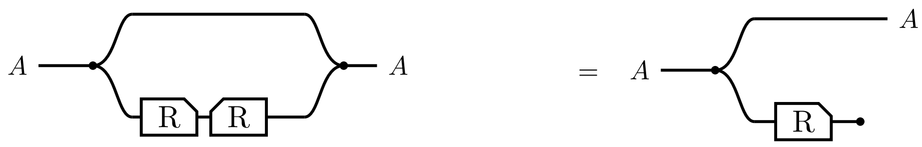
The last of Freyd's §2.21 equations that have no picture here — a tape around nothing, or a tape around what is already inside it, is not worth drawing. `R` = people who admire `a`, `S` = people who admire `b` throughout.

equation	the reading
`R ∪ (S ∩ R) = R` `(R ∪ S) ∩ R = R` absorption	People who admire `a`, or who admire both: still the people who admire `a`. People who admire `a` or `b`, and who admire `a`: the same. The smaller side of each pair is already inside the larger.

equation	the reading
`R ⊥ = ⊥` `R 0_S = 0_{R S}` — and `⊥ R = ⊥` on the other side	Admiring someone and then taking a step that leads nowhere gets you nowhere. `⊥` is a two-sided zero for composition, as `⊤` is not.

§6. Domain and range

> The **domain** `Dom R ≜ 𝟙 ∩ R R°` and the **range** `Ran R ≜ Dom R°`.



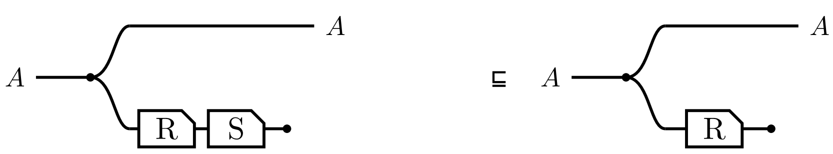
▷ lands the return leg back on the value ◁ handed out,
so R° ▷ cuts to ⊸ — Frobenius

Running `R` and throwing the result away leaves only the fact that `R` could fire, and `Ran R` is the same picture with the box mirrored. In `Rel` both steps are `{(a,a) : ∃b. a R b}`.

`Dom R ≜ 𝟙 ∩ R R°`
`Dom R ⊑ A` ⟺ `R ⊑ A R`, for `A` coreflexive
`Dom (R S) ⊑ Dom R`
`Dom (R ∩ S) = 𝟙 ∩ S R°`
`R` entire ⟺ `Dom R = 𝟙` ⟺ `𝟙 ⊑ R R°`
`R` simple ⟺ `R° R ⊑ 𝟙`
`R` a map ⟺ `R` entire and simple
`R, S` entire ⟹ `R S` entire — likewise simple, likewise maps
`R S` entire ⟹ `R` entire

§6.1. Sliding the discard

`Dom (R S) ⊑ Dom R`, and a single glyph for `Dom` would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

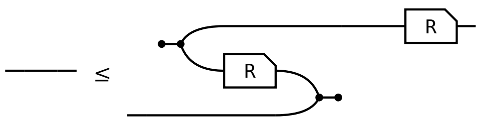
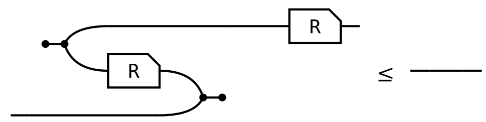
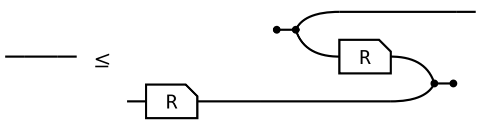
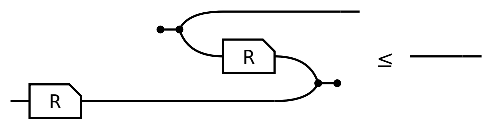


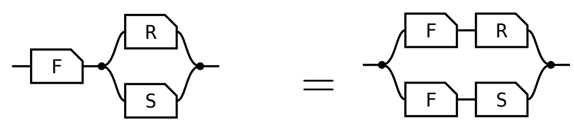
`S ⊸ ⊑ ⊸`, the lax axiom for ⊸ in the first section
— the discard slides back past `S`

Equality is `S` entire, which is the same picture read as `Dom R = 𝟙 ⟺ R` entire.

§7. Maps

Every arrow is lax for ◁ and for ⊸ by axiom, and for ▷ and ⟜ by taking ° of those two. All four hold for **every** arrow, their REVERSES do not, and each reverse holding is a property of the arrow. The right-hand column is the **adjoint** form, which is the definition used here because it is literally the allegory's `Simple` and `Entire`; that the two forms agree is a separate theorem, not proved here.

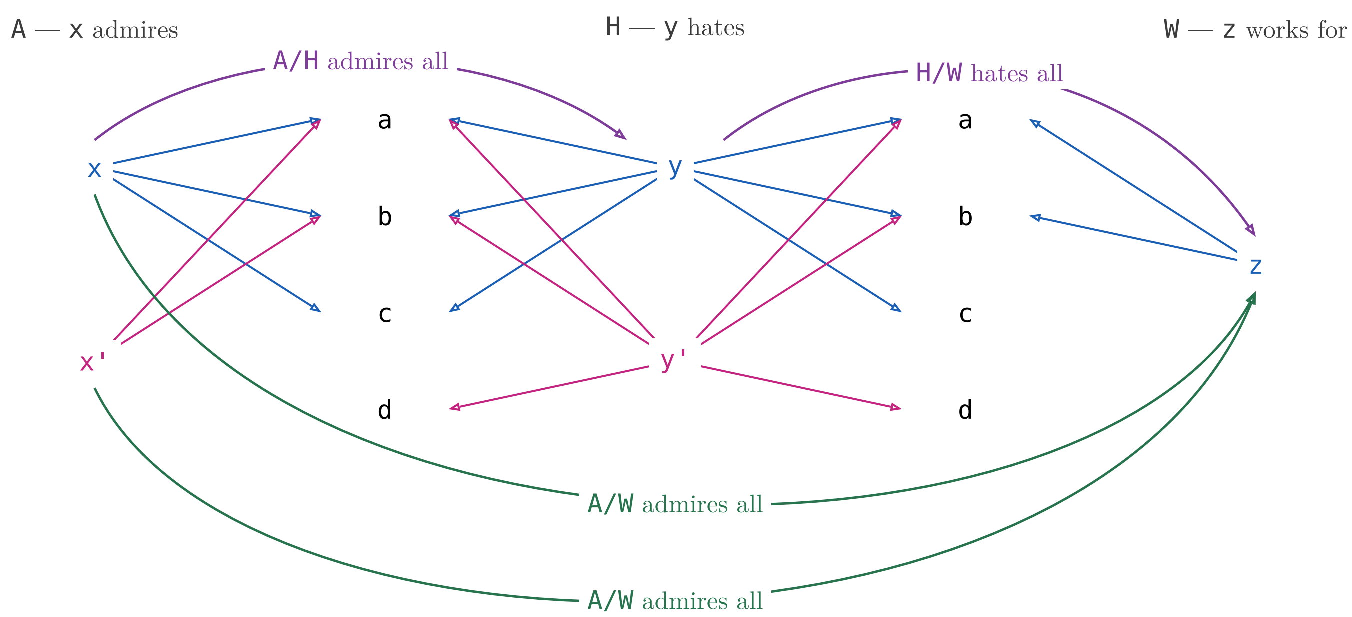
 **surjective** $\mathbb{1} \sqsubseteq R^\circ\ R$, that is, R° entire.	 **single valued** $R^\circ\ R \sqsubseteq \mathbb{1}$
 **entire** $\mathbb{1} \sqsubseteq R\ R^\circ$. With **single valued**, a **map**.	 **injective** $R\ R^\circ \sqsubseteq \mathbb{1}$

law	picture
`F (R ∩ S) = F R ∩ F S` F single valued F = people x may ask, R = admires a, S = admires b. Single valued means one person, so the sides agree.	 one person who admires both a at A, b at B

§8. / is all of

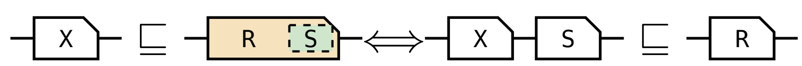
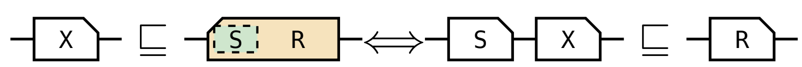
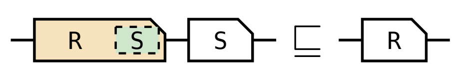
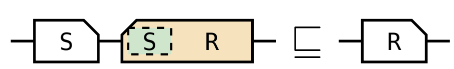
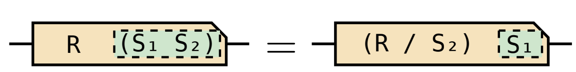
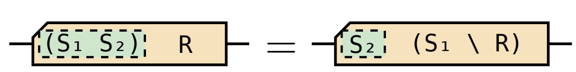
```
x (R/S) y  ⟺  ∀p. y S p → x R p
x (S\R) y  ⟺  ∀p. p S x → p R y
/ compares images: S(y) ⊆ R(x).   \ compares preimages: S°(x) ⊆ R°(y).
example: A admires, H hates, W works for
x (A/H) y — x admires everyone y hates.
x (H\A) y — everyone who hates x admires y.
```

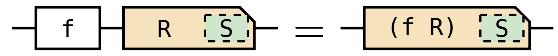
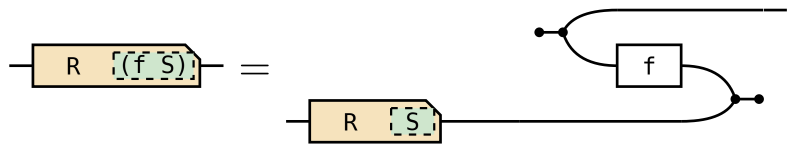
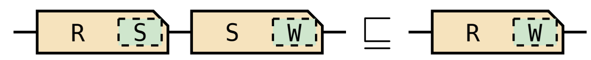
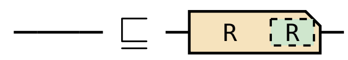
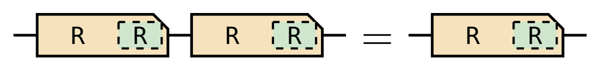
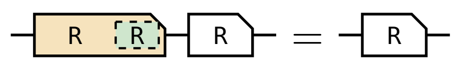
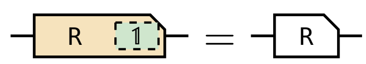
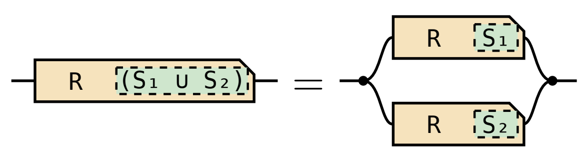
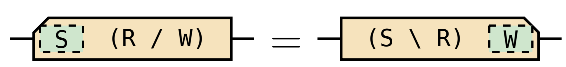
§8.1. $(R/S)(S/W) \sqsubseteq R/W$



x (A/H) y — x admires everyone y hates
y (H/W) z — y hates everyone z works for
x (A/W) z — x admires everyone z works for

(A/H)(H/W) is a path: x → y → z, and that is all of it. A/W also holds of x', who admires everyone z works for — but x' does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of $(R/S)(S/W) \sqsubseteq R/W$.

$X \sqsubseteq R/S \iff X\ S \sqsubseteq R$ X is any x-to-y pairing; one that only pairs x with a y such that x admires everyone y hates lies inside R/S, and R/S is the largest such.	
$X \sqsubseteq S\backslash R \iff S\ X \sqsubseteq R$ The mirror — divide on the left when X comes first.	
$(R/S)\ S \sqsubseteq R$ There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at $S = \varnothing$: R/S is everyone, $(R/S)\ S = \varnothing$.	
$S\ (S\backslash R) \sqsubseteq R$ The mirror.	
associate: $R/(S_1\ S_2) = (R/S_2)/S_1$ **A friend's enemy** is two hops: divide by the far end first.	
$(S_1\ S_2)\backslash R = S_2\backslash(S_1\backslash R)$ The mirror.	

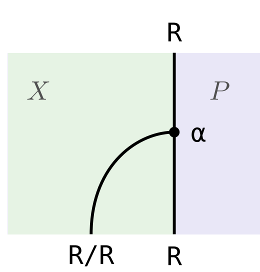
maps: `f (R/S) = (f R)/S` Rename `x` before or after dividing — the licence to write `f R / S`.	
`R/(f S) = (R/S) f°` Rename `y`: a map leaves a denominator as `f°` outside the box.	
`(R/S)(S/W) ⊑ R/W` Someone who admires all of a hate-list that already covers everyone `z` works for admires those people too.	
`𝟙 ⊑ R/R` `R/R` runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only `a` and `b` admire each other's idols too, and still stay two people.	
`(R/R)(R/R) = R/R` `R/R` is the preorder **admires at least as much as**, and a preorder is idempotent. Freyd writes ⊑; with `𝟙 ⊑ R/R` above it is an equality.	
`(R/R) R = R` Reaching `p` through someone whose idols `x` fully admires is reaching `p` directly, since `x` admires their own idols.	
`R/𝟙 = R` Dividing by `𝟙`: `p`'s list is just `{p}`, so admiring all of it is admiring `p`.	
`R/(S₁ ∪ S₂) = R/S₁ ∩ R/S₂` Admiring a combined hate-list is admiring each list in full.	
`S\(R/W) = (S\R)/W` Which is why `S\R/W` needs no bracket.	

Fifteen laws, fifteen pictures, and not one shows a generator: `∩`, `∪`, `°` and composition are what the Frobenius generators build, and `/` is none of those — it is posited, with nothing to unfold.

§8.2. `R/R` is a preorder, which is what a monad is here

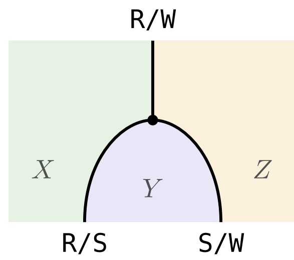
A **monad** in a locally posetal 2-category is a 1-cell `M` with `𝟙 ⊑ M` and `M M ⊑ M` — a unit and a multiplication, and with the hom-posets thin there is nothing else to give. In `Rel` that is a reflexive transitive relation: a preorder. The table above proves both halves at `M := R/R`, and names the preorder they define.

`(R/R) R = R` is then the action that makes `R` a module over that monad — Hinze and Marsden's $\alpha : M \circ A \dot{\to} A$ (IntroString p. 83, laws (3.10a) and (3.10b)) at M := `R/R`, A := `R`. Their two coherence conditions are free here: parallel 2-cells are unique, so any two of them are equal.



Green is `X`, where `x` lives; purple is `P`, where the `p` of the definition lives. `R/R : X → X` runs inside `X` and ends on `R : X → P` at the action `α`, which is `(R/R) R ⊑ R`.

§8.3. The whole family is one preorder



The same dot with one region more: `R/S : X → Y` and `S/W : Y → Z` come in, `R/W : X → Z` goes out. Green `X`, purple `Y`, amber `Z`.

Closing `Y` off is the mid-point `y` being quantified away; the `⊑` rather than `=` is the loss the figure for that law above already draws.

Concretely in `Rel`, `𝟙 ⊑ R/R` and `(R/S)(S/W) ⊑ R/W` are one preorder on the disjoint union of `X`, `Y`, `Z`, …: reflexivity is `𝟙 ⊑ R/R` inside a block, transitivity is `(R/S)(S/W) ⊑ R/W` across blocks.

Abstractly the same two are a **lax functor** out of the codiscrete category on `{R, S, W, …}` — one object per relation, exactly one arrow each way — sending `R` to `X`, the object its rows are indexed by, and the arrow `R → S` to `R/S : X → Y`. That is Bénabou's **polyad** (Bénabou 1967, Def. 5.5), of which a monad is the one-object case: the subsection above is this one at `R = S = W`.

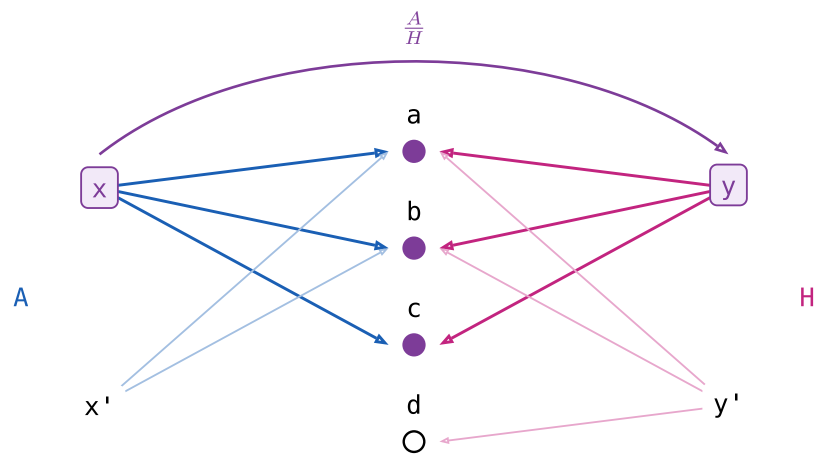
`R/S` is at the same time a bimodule between the two monads `R/R` and `S/S`: `(R/R)(R/S) ⊑ R/S` and `(R/S)(S/S) ⊑ R/S`, each the counit `(R/S) S ⊑ R` with one action whiskered on — `(R/R) R = R` on the left, `(S/S) S = S` on the right.

§9. `x (Admires%Hates) y` is `x admires only all whom y hates`

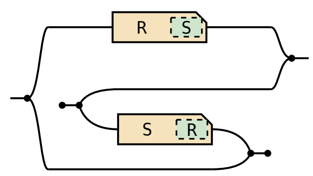
> $\frac{R}{S} \triangleq$ `(R/S) ∩ (S/R)°`. In `Rel` `x` and `y` has the same image: `∀p. (x R p ⟺ y S p)`

`x (A/H) y` — `x` admires everyone `y` hates
`x ((H/A)°) y` — `x` admires only people `y` hates
`x ((A/H) ∩ (H/A)°) y` — `x` admires only and all whom `y` hates

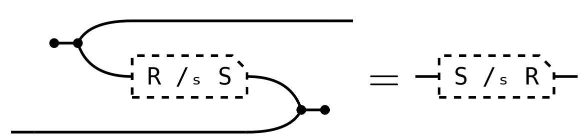
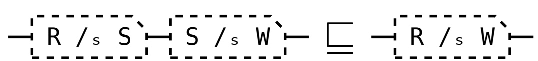
`x` admires exactly whom `y` hates: `/` is **all of**, $\frac{R}{S}$ is **only all of**.



A meet of two long divisions, the second turned round by the converse frame:



exported from the definition, not transcribed

$\left(\frac{R}{S}\right)^{\circ} = \frac{S}{R}$ Matching is symmetric.	
$\frac{R}{S}\frac{S}{W} \sqsubseteq \frac{R}{W}$ And transitive.	

No pictures for the rest of §2.35: symmetric division is not built from the generators.

law	**the reading**
$X \sqsubseteq \frac{R}{S} \iff XS \sqsubseteq R$ and `X° R ⊑ S`	`X` may pair `x` with `y` only when `x` admires exactly whom `y` hates. Both halves must typecheck, so the operation is **partial**.
$\frac{R}{S}S \sqsubseteq R$	$\left(\exists y.\, x\left(\frac{R}{S}\right)y \wedge ySp\right) \to xRp$ `x only admires whom y hates` $\frac{R}{S}S = \mathrm{Dom}\left(\frac{R}{S}\right)R$

$\frac{R}{R}R = R$	$\left(\exists y.\, x\left(\frac{R}{R}\right)y \wedge yRp\right) \iff xRp$
	 `x and y admire the same people`
	 `y = x always qualifies (𝟙 ⊑ R%R below)`

$\mathbb{1} \sqsubseteq \frac{R}{R}$	$x\left(\frac{R}{R}\right)y$ if `x` and `y` admires the same peoples.
$\left(\frac{R}{R}\right)^2 = \frac{R}{R}$	So the relation **admires the same people** is an equivalence relation.
$X \sqsubseteq \frac{R}{R} \iff XR \sqsubseteq R$, for symmetric `X`	The largest symmetric arrow that leaves `R` alone.

law	the reading
$\frac{R}{\mathbb{1}}$ is the **simple part** of `R`	The people who admire exactly one person and nobody else. It equals `R` only when `R` is simple, unlike `R/𝟙 = R`.
`Dom` $\frac{R}{S}$ `= 𝟙 ∩ (R/S)(S/R)`	Its domain is the **domain of simplicity** of `R`.

§9.1. Straight

> `S` is **straight** when $\frac{S}{S} = \mathbb{1}$ — no two `y`s hate the same people.

law	the reading
every symmetric `X` with `X S ⊑ S` is coreflexive	Equivalently, `S` is straight.
`f S = g S ⟹ f = g`	A straight `S` tells its `y`s apart, so it cancels on the right.
`S R` straight ⟹ `S` straight	If the longer chain tells them apart, the first step already does.
`S` straight ⟺ `R/S` simple for all `R`	If no two `y`s agree, at most one `x` can match a given `y`.
`R = h S`, `h` a cover, `S` straight	In an effective division allegory every arrow factors that way.

§10. Power allegories

> A **power allegory** is a division allegory with one unary operation on arrows, ∋ **epsiloff**, subject to
>
> $$\ni_R\Box = R\Box, \quad \ni_R = \ni_{R\,\Box}$$
> $$\mathbb{1} \sqsubseteq (R/\ni_R)(\ni_R/R) \qquad \ni_R \text{ is } \textbf{thick}$$
> $$(\ni_R/\ni_R) \cap (\ni_R/\ni_R)^\circ \sqsubseteq \mathbb{1} \qquad \ni_R \text{ is } \textbf{straight}$$
>
> `R □` is `R`'s target, an identity arrow. For `R : A → B` write `∋ : [B] → B`, dropping the subscript.

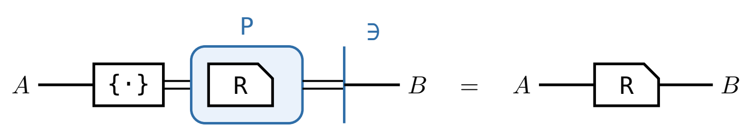
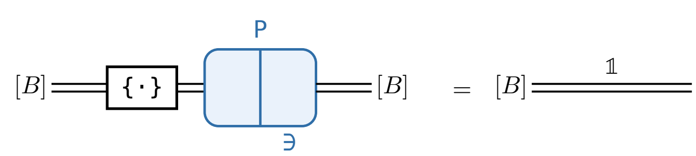
In `Rel` its source is the powerset of its target and `l ∋ i` iff `i ∈ l`, and `∋` reads a list `l` back into the people on it.

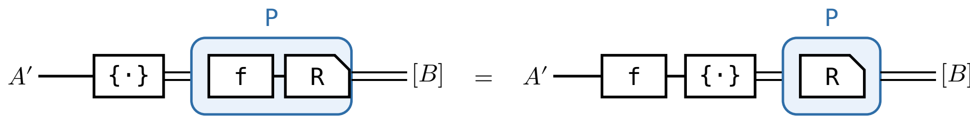
> `Λ(R)` ≜ $\frac{R}{\ni}$ `: A → [B]`, for `R : A → B`. The list-of map: send `x` to the list of the people `x` admires.

One convention: a `[B]`-wire is drawn double, because it is a `B`-wire one level down — inside the box. The box is the power relator `P`, which the relator section below defines: a box holding `R : B → C` is `P R : [B] → [C]`. Inside the box lives the allegory, outside only maps, and `i : Map(𝓐) ↪ 𝓐` is the inclusion that forgets the difference. The adjunction `i ⊣ P`, that is `𝓐(A, B) ≅ Map(A, [B])`, has the two ends:

counit: open the box — unit: the one-person list

`Λ` is neither of them — it is the transpose the adjunction gives, `Λ(R) = {·} (P R)`: make the singleton, then push `R` through the box.

law	the reading
$\ni_R\Box = R\Box, \quad \ni_R = \ni_{R\,\Box}$	`∋` has the same target as `R`, and replacing `R` by the identity at that target leaves it unchanged: one `∋` per object, not per arrow.
`∋` is **thick**	**Comprehension**: every `x` has a list of exactly the people `x` admires. Equivalently every `R` factors as a map followed by `∋`.
`∋` is **straight**, that is $\frac{\ni}{\ni} \sqsubseteq \mathbb{1}$	**Extensionality**: two lists with the same people are the same list.
`Λ(R)` is simple	`∋` is straight, and dividing by a straight arrow is simple. At most one list per `x`.
`Λ(R)` is entire ⟺ `∋` thick	`Dom` $\frac{R}{\ni}$ `= 𝟙 ∩ (R/∋)(∋/R)`, the domain row above. At least one list per `x`, so `Λ(R)` is a **map**.
`Λ(R) ∋ = R` Everything left of the `∋` wall is `Λ(R)`; open the wall and what was inside comes out unchanged. That is the β of this calculus.	
`Λ(R)` is the only map with `Λ(R) ∋ = R`	Two maps naming the same people name the same list — extensionality again.
`F ⊑ Λ(F ∋)`, `F` simple	A partial choice of lists is inside the total one.
`[α]` = source of `∋`, the **power-object**	Every list over `α`.
`{·} ≜ Λ(𝟙)`, the **singleton map**, monic	The one-person list. `Λ(𝟙)Λ°(𝟙) ⊑` $\frac{\mathbb{1}\,\ni}{\ni\,\mathbb{1}} \sqsubseteq \frac{\mathbb{1}}{\mathbb{1}} \sqsubseteq \mathbb{1}$.
`Λ(∋) = 𝟙` Make the list of a list, then read it back one level down. Straightness itself, and one of the two triangle identities of `i ⊣ P`; the other is `{·} ∋ = 𝟙`.	

law	the reading
fusion: `Λ(f R) = f Λ(R)`, `f` a map `f` is a plain rectangle — a map, so it may cross the `{·}` node. That is naturality of the unit, `f {·} = {·} (P f)`, and it is the whole content of fusion; a chamfered box is stuck on the side it is on.	
`Λ(f) = f {·}`, `f` a map	Rename first or take singletons first — the fusion row above at `R = 𝟙`.
$\frac{R}{S} = \Lambda(R)\Lambda^\circ(S)$ `x` and `y` match exactly when the two boxes hold one and the same list — the `[B]`-wire joining them is the whole statement.	
`C` a topos ⟹ `Rel(C)` a power allegory	And back: a unitary tabular power allegory has `Map(A)` a topos. Extensionality and comprehension are all a topos adds.

§11. Relator

> Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from R to S IS $R \sqsubseteq S$. A **relator** $F : \mathcal{C} \to \mathcal{D}$ is a 2-functor between allegories:
>
> $$F\ \mathbb{1} = \mathbb{1} \qquad F(R\ S) = (F\ R)(F\ S) \qquad R \sqsubseteq S \implies F\ R \sqsubseteq F\ S$$
>
> Preserving $^\circ$ is **not** asked for — $^\circ$ is an identity-on-objects involution $\mathcal{C}^{op} \to \mathcal{C}$, no part of the 2-category.

- For f a map, $F\ f$ is a map and $F(f^\circ) = (F\ f)^\circ$.
- Over a **tabular** allegory a functor is a relator $\iff$ it preserves $^\circ$.
- A relator is fixed by what it does to maps.
- $F(R \cap S) \sqsubseteq (F\ R) \cap (F\ S)$, and strictly.
- $F(X \cap Y) = (F\ X) \cap (F\ Y)$ for $X,\ Y$ coreflexive.
- $F(\mathrm{Dom}\ R) = \mathrm{Dom}(F\ R)$ for F preserving $^\circ$.

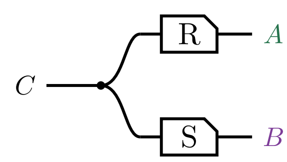
The **power relator** P — $x\ (P\ R)\ y \iff (\forall a \in x.\ \exists b \in y.\ a\ R\ b) \wedge (\forall b \in y.\ \exists a \in x.\ a\ R\ b)$ — is where the fourth is strict: for $R = \{(a_1,b_1),\ (a_2,b_2)\}$ and $S = \{(a_1,b_2),\ (a_2,b_1)\}$ the pair $(\{a_1,a_2\},\ \{b_1,b_2\})$ is in $P\ R \cap P\ S$, while $R \cap S = \varnothing$.

§11.1. The pairing $\langle R,S\rangle$

> The **pairing** of $R : C \to A$ and $S : C \to B$ is $\langle R,S\rangle \triangleq R\ \pi_1^\circ \cap S\ \pi_2^\circ$, where $(\pi_1,\ \pi_2)$ is the tabulation of $\top$.

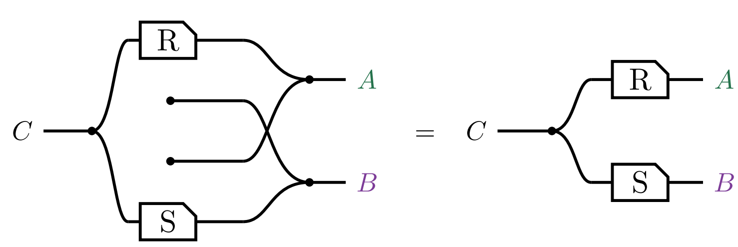
$$\langle R,S\rangle\ \pi_1 = (\mathrm{Dom}\ S)\ R \qquad \langle R,S\rangle\ \pi_2 = (\mathrm{Dom}\ R)\ S$$

$$\begin{tikzcd}
 & A & \\
C \arrow[ur, "R"] \arrow[dr, "S"'] \arrow[rr, dashed, "{\langle R,S\rangle}"] & & A \times B \arrow[ul, "\pi_1"'] \arrow[dl, "\pi_2"] \arrow[ul, phantom, "\sqsupseteq" near start] \arrow[dl, phantom, "\sqsupseteq" near start] \\
 & B &
\end{tikzcd}$$



A domain is coreflexive, so $\langle R,S\rangle\ \pi_1 \sqsubseteq R$, with equality exactly when S is entire; for maps both triangles commute and $\langle f,g\rangle$ is unique. In Rel, $c\ \langle R,S\rangle\ (a,b)$ iff $c\ R\ a$ and $c\ S\ b$ — copy c, then R on one strand and S on the other, which is $\lhd\ (R \otimes S)$ on the right.

No $^\circ$ survives the translation. $\pi_1 = \mathbb{1} \otimes ⊸$ discards the second component, so $\pi_1^\circ = \mathbb{1} \otimes ⟜$ **creates** one out of nothing, and $\cap$ is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:

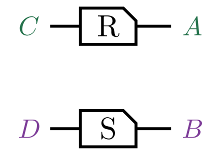


⟜ $\rhd$ = $\mathbb{1}$ on each half

§11.2. The relational product $R \times S$

> $R \times S \triangleq \langle \pi_1\ R,\ \pi_2\ S\rangle$, a relator in each argument but no longer a categorical product.

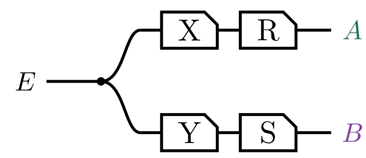
$$\begin{tikzcd}
C \arrow[rr, "\mathsf{R}"] & & A \\
C \times D \arrow[u, "\pi_1"] \arrow[d, "\pi_2"'] \arrow[rr, dashed, "R \times S"] \arrow[urr, phantom, "\sqsupseteq"] \arrow[drr, phantom, "\sqsupseteq"] & & A \times B \arrow[u, "\pi_1"'] \arrow[d, "\pi_2"] \\
D \arrow[rr, "\mathsf{S}"] & & B
\end{tikzcd}$$



Right-then-up is `(R × S) π₁`, up-then-right is `π₁ R`, and `(R × S) π₁ ⊑ π₁ R`, equality when `S` is entire. In `Rel`, `(c,d) (R × S) (a,b)` iff `c R a` and `d S b` — two strands side by side, no copy dot: `R × S = R ⊗ S`.

§11.3. Absorption

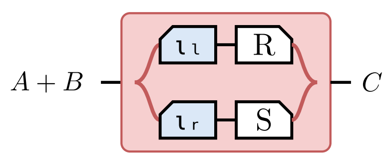
For `X : E → C` and `Y : E → D`, `⟨X,Y⟩ (R × S) = ⟨X R, Y S⟩`. Both sides are this picture:



§11.4. The coproduct `[R,S]`

The injections `ιₗ : A → A + B` and `ιᵣ : B → A + B` are maps, and the coproduct they make of the maps stays a coproduct once every arrow is allowed: both equations hold with equality and `[R,S]` is the only arrow satisfying them, with none of the `Dom` slack `⟨R,S⟩` carries.

> `[R,S] ≜ ιₗ° R ∪ ιᵣ° S`, and `R + S ≜ [R ιₗ, S ιᵣ]`.



The tape is the union — a particle entering at `A + B` takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.

`ιₗ [R,S] = R`, `ιᵣ [R,S] = S`, and `[R,S]` is the only such arrow
`ιₗ ιₗ° = 𝟙 = ιᵣ ιᵣ°`
`ιₗ ιᵣ° = ⊥ = ιᵣ ιₗ°`
`ιₗ° ιₗ ∪ ιᵣ° ιᵣ = 𝟙`
`[U,V]° [R,S] = U° R ∪ V° S`

The first row is not free: `ιₗ, ιᵣ` were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because `Λ` sends an arrow `A → C` to a map `A → [C]` reversibly, so the map coproduct can be applied underneath it. For any `T : A + B → C`,

$$\begin{aligned}
& \iota_l\ T = R \text{ and } \iota_r\ T = S \\
\iff\ & \quad \text{Λ is injective — } Λ(X)\ \ni\ =\ X \text{ gives } X \text{ back} \\
& Λ(\iota_l\ T) = Λ(R) \text{ and } Λ(\iota_r\ T) = Λ(S) \\
\iff\ & \quad \text{Λ fusion, } Λ(f\ X)\ =\ f\ Λ(X) \text{ for } f \text{ a map} \\
& \iota_l\ Λ(T) = Λ(R) \text{ and } \iota_r\ Λ(T) = Λ(S) \\
\iff\ & \quad \text{the coproduct of maps, at the map } Λ(T) \\
& Λ(T) = [Λ(R),\ Λ(S)] \\
\iff\ & \quad Λ(X) \text{ is the only map with } Λ(X)\ \ni\ =\ X \\
& T = [Λ(R),\ Λ(S)]\ \ni
\end{aligned}$$

Every step is an $\Longleftrightarrow$, so `[Λ(R), Λ(S)] ∋` is the **only** arrow satisfying the two equations — which is what `[R,S]` was claimed to be. Drawn:

$$
\begin{tikzcd}
 & A \arrow[dl, "\iota_l"'] \arrow[dr, "\Lambda(R)"] \arrow[drr, bend left, "R"] & & \\
A + B \arrow[rr, dashed, "{[\Lambda(R),\ \Lambda(S)]}"] & & {[C]} \arrow[r, "\ni"] & C \\
 & B \arrow[ul, "\iota_r"] \arrow[ur, "\Lambda(S)"'] \arrow[urr, bend right, "S"'] & &
\end{tikzcd}
$$

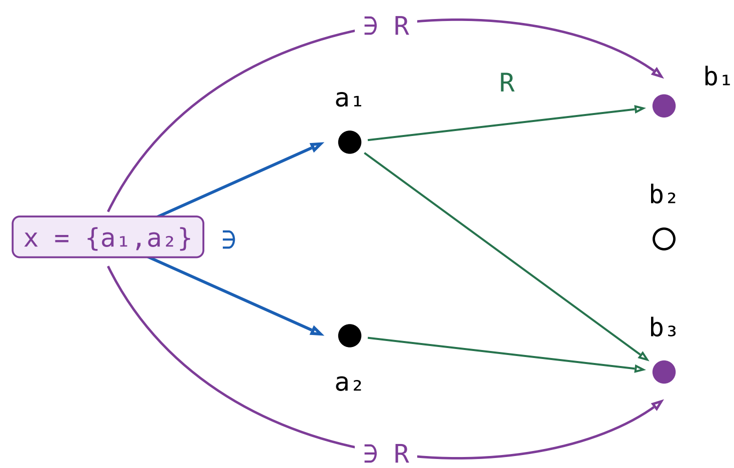
Nothing here holds only up to `⊑`: every triangle commutes on the nose, which is the difference from the pairing above. The border spells `[R,S] = [Λ(R), Λ(S)] ∋`, and pushing `∋` into the union that `[·,·]` on maps already is turns that back into the definition, $(\iota_l^\circ\ \Lambda(R) \cup \iota_r^\circ\ \Lambda(S))\ \ni\ = \iota_l^\circ\ R \cup \iota_r^\circ\ S$.

§11.5. The power relator `P R`

> For `R : A → B`, `P R ≜ ((∋ R)/∋) ∩ ((∋ R°)/∋)° : [A] → [B]`.

`∋ R` is a **composition** — `∋ : [A] → A` followed by `R` — and not Freyd's subscripted $\ni_R$, which is the `∋` at `R`'s **target**: §11 writes one `∋` per object and drops the subscript. Piece by piece, on one instance, with `x, y` lists and `a, b` their elements.

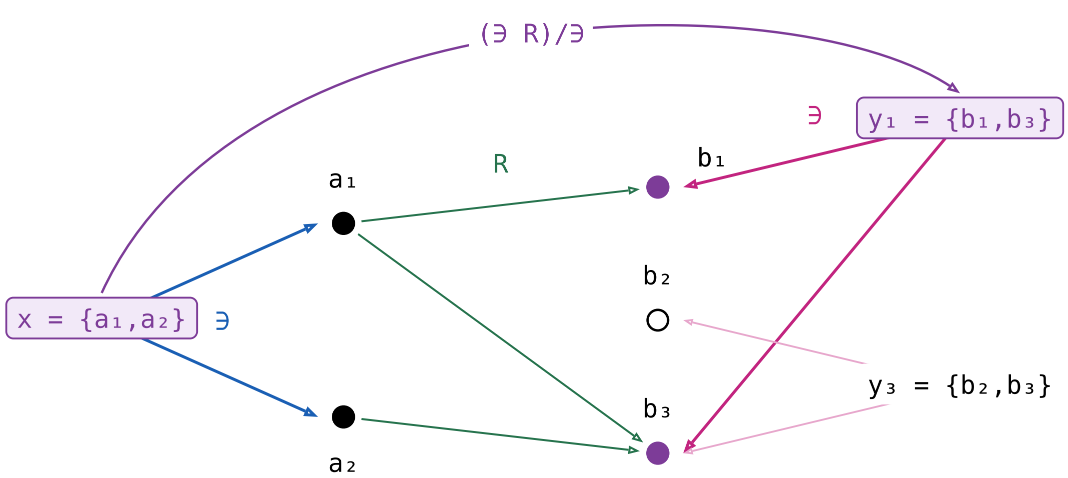
$x = \{a_1, a_2\}$, and `R` sends a_1 to b_1 and b_3, a_2 to b_3 alone. Nothing on `x` reaches b_2, so b_2 is drawn hollow.



`∋ R : [A] → B`   `x (∋ R) b ⟺ ∃a ∈ x. a R b`

one arc per element of `B` that `x` reaches, and b_2 gets none

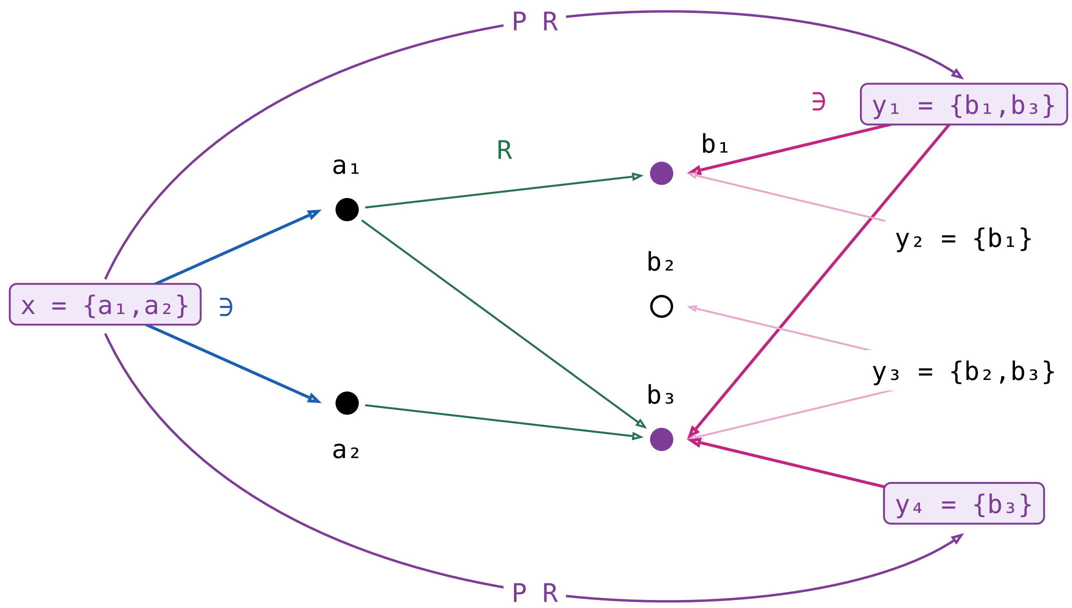
Dividing by `∋` turns that into a relation between **lists**: every element of `y` must be one of the filled dots.



`(∋ R)/∋ : [A] → [B]`   `x ((∋ R)/∋) y ⟺ ∀b ∈ y. ∃a ∈ x. a R b`

y_3 is rejected: it names b_2, which `x` does not reach

The other half of the meet is that clause with the lists swapped and R turned round: every element of x must reach y. Both together, and two more lists to separate them:



$((\ni R^\circ)/\ni)^\circ : [A] \to [B] \qquad x\ ((\ni R^\circ)/\ni)^\circ\ y \iff \forall a \in x.\ \exists b \in y.\ a\ R\ b$

y_2 is rejected too: a_2's only image is b_3, which y_2 does not name

In words, if $x\ (P\ R)\ y$ then every element of x is related by R to some element of y, and conversely — the two clauses of ∩, one per direction.

Neither ∋ nor P R is a map. ∋ sends a list to **each** of its elements; P R sends x to **every** y meeting the two clauses, y_1 and y_4 both. Nor is it entire: an $a \in x$ with no image at all leaves x with no partner whatever. Turning a relation into a map is Λ's job (§11), and on lists that map is $\Lambda(\ni R)$ — of the four it accepts y_1 only.

Not a symmetric division. $\frac{\ni R}{\ni}$ is $\Lambda(\ni R)$, and in Rel it reads $x\ \Lambda(\ni R)\ y \iff y = \{b \mid \exists a \in x.\ a\ R\ b\}$: in words, y is **exactly** what x reaches, where P R asks only that each side cover the other. Nor can it be repaired: in that fraction's second half $(\ni / (\ni R))^\circ$ the R sits in a denominator, so the operation is antitone there, and a relator has to be monotone. P R keeps the first half and takes the second half at R°, which puts R back in a numerator. The fraction returns exactly where the two halves agree — at 𝟙, and at a map.

law	the reading
$X \sqsubseteq P\ R \iff X \ni\, \sqsubseteq\, \ni R$ and $X^\circ \ni\, \sqsubseteq\, \ni R^\circ$	One containment, and the same one at R° — which is the definition read off the two divisions. Hence $P(R^\circ) = (P\ R)^\circ$, and $R \sqsubseteq S \implies P\ R \sqsubseteq P\ S$.
$P\ 𝟙 = \frac{\ni}{\ni} = 𝟙$	The straightness axiom verbatim: extensionality **is** P's unit law.
$P\ f = \Lambda(\ni f) = \frac{\ni f}{\ni}$, for f a map	In Rel, $x\ (P\ f)\ y \iff y = \{f\ a \mid a \in x\}$. The half at f° says every $a \in x$ has its $f\ a$ on y; f has just the one image per a, so that already says y contains everything x reaches, which is the fraction's second half. For a map the two definitions coincide.
$P(R\ S) = (P\ R)(P\ S)$	⊒ is the division cancellation laws. ⊑ is the one law in this section that is not a calculation: it needs a tabulation of $P(R\ S)$.

§12. Catamorphism

> F a relator with an **initial algebra** $\alpha : F\ T \to T$ among the maps. For a relational algebra $R : F\ B \to B$, the **catamorphism** $\llparenthesis R \rrparenthesis : T \to B$ is the unique arrow with $\alpha\ \llparenthesis R \rrparenthesis = (F\ \llparenthesis R \rrparenthesis)\ R$.

$$\begin{tikzcd}
F\ T \arrow[r, dashed, "F\ X"] \arrow[d, "\alpha"'] & F\ B \arrow[d, "R"] \\
T \arrow[r, dashed, "X"'] & B
\end{tikzcd}$$

With $X = \llparenthesis R \rrparenthesis$ the square commutes strictly: unlike the product's triangles, there is no slack here, so no $\sqsubseteq$ appears anywhere in it.

The three **fusion** rows rewrite $\llparenthesis R \rrparenthesis\ S$ through a second algebra $Q : F\ C \to C$ along an arrow $S : B \to C$. They are the only rows here that need the allegory **locally complete** — every hom-set a complete lattice — because they come from a least-fixed-point argument; the rest of the table needs only the initial algebra.

name	law	what it says
the defining equation	$X = \llparenthesis R \rrparenthesis \iff \alpha\ X = (F\ X)\ R$	Folding a value that α has just built is the same as folding its parts and combining them with R. Nothing else has that property.
Lambek	$\alpha^{\circ} = \alpha^{-1}$, the initial algebra is an isomorphism	A constructor can be undone — α° takes a value apart into the parts it was built from.
Eilenberg–Wright	$\Lambda\llparenthesis R \rrparenthesis = \llparenthesis \Lambda((F\ \ni)\ R) \rrparenthesis$	A relational fold is a deterministic fold of SETS: Λ pushes the nondeterminism into the power-object, where the fold is a map again.
Eilenberg–Wright	$\llparenthesis R \rrparenthesis = \llparenthesis \Lambda((F\ \ni)\ R) \rrparenthesis\ \ni$	The same fact read back — fold deterministically into a set, then take a member of it.
fusion	$\llparenthesis Q \rrparenthesis \sqsubseteq \llparenthesis R \rrparenthesis\ S \Longleftarrow (F\ S)\ Q \sqsubseteq R\ S$	Half of fusion: an inclusion between the two algebras is inherited by the folds.
fusion	$\llparenthesis R \rrparenthesis\ S \sqsubseteq \llparenthesis Q \rrparenthesis \Longleftarrow R\ S \sqsubseteq (F\ S)\ Q$	The other half, with both inclusions turned around.
fusion	$\llparenthesis R \rrparenthesis\ S = \llparenthesis Q \rrparenthesis \Longleftarrow R\ S = (F\ S)\ Q$	Both halves at once: a fold followed by S collapses into a single fold, which is how an intermediate structure is got rid of.

The two Λ rows, drawn:

$$\begin{tikzcd}
F\ T \arrow[r, dashed, "F\ K"] \arrow[d, "\alpha"'] & F\ [B] \arrow[r, "F\ \ni"] \arrow[d, "\Lambda((F\ \ni)\ R)"] & F\ B \arrow[d, "R"] \\
T \arrow[r, dashed, "K"'] & [B] \arrow[r, "\ni"'] & B
\end{tikzcd}$$

The left square is that catamorphism's own defining square, $K = \llparenthesis \Lambda((F\ \ni)\ R) \rrparenthesis$, and the right one is Λ's cancellation, so the outer rectangle says $K\ \ni$ satisfies the defining equation of $\llparenthesis R \rrparenthesis$ — and uniqueness finishes it.

§13. Type functor

> `F` a **binary** relator: `F(R, S)` is its action on a pair, and `F X` abbreviates `F(𝟙, X)`, the `F` of the catamorphism section. For every object `A` the initial algebra is `α : F(A, T A) → T A`, among the maps. The **type functor** `T` acts on an arrow `R : A → B` by
>
> `T R = ⦇F(R, 𝟙) α⦈ : T A → T B`

`F` is the **base functor** — one layer of the structure, acting on the recursive position — while `T` is the datatype itself, acting on the parameter. For cons-lists, `list R = ⦇[nil, (R ⊗ 𝟙) cons]⦈`, which is `map R`.

$$
\begin{tikzcd}
F(A, T A) \arrow[r, dashed, "F(\mathbb{1}, T R)"] \arrow[d, "\alpha"'] & F(A, T B) \arrow[r, "F(R, \mathbb{1})"] & F(B, T B) \arrow[d, "\alpha"] \\
T A \arrow[rr, dashed, "T R"'] & & T B
\end{tikzcd}
$$

`T R` is the unique arrow making it commute; there is no `⊑` in it.

name	law	what it says
the defining equation	`T R = ⦇F(R, 𝟙) α⦈`	Rebuild the structure with `α`, applying `R` to the parameter on the way; that is `map R`.
functor	`T 𝟙 = 𝟙` and `(T R)(T S) = T (R S)`	Mapping the identity changes nothing, and two maps in a row are one map.
type functor fusion	`(T R) ⦇Q⦈ = ⦇F(R, 𝟙) Q⦈`	A map followed by a fold is a single fold — the mapped structure is never built.
naturality of `α`	`α (T R) = F(R, T R) α`	Building and then mapping is the same as mapping the parts and then building, so `α` is natural from `G R = F(R, T R)` to `T`.
type relator	`(T R)° = T (R°)`	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

Type functor fusion is the equality fusion row above applied to `T R`'s own defining algebra, whose side condition holds because `F` is a bifunctor — `F(R, 𝟙) F(𝟙, ⦇Q⦈) = F(R, ⦇Q⦈) = F(𝟙, ⦇Q⦈) F(R, 𝟙)` — so it inherits that row's local-completeness requirement.

$$
\begin{tikzcd}
F(A, T A) \arrow[r, "F(R, T R)"] \arrow[d, "\alpha"'] & F(B, T B) \arrow[d, "\alpha"] \\
T A \arrow[r, "T R"'] & T B
\end{tikzcd}
$$

It commutes strictly: it is the naturality square of `α`.